

Programming in C++

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Q1: Manipulators:

Manipulators are special functions in C++ that help in formatting the output on the console.

They are used along with cout and cin to change the way data is displayed on read.

Q2: Why are they used?

Manipulators makes the output more organized and user-friendly.

They are used to:

- Sets the width of output fields. (setw).
- Move to the next line (endl).
- Change the display style of floating numbers (fixed, scientific).

- Control the no. of digits shown after the decimal point. (set precision).

In short, it gives control over how data looks when printed.

Q3: Write and run the example programs given below?

Commonly used Manipulators:

- Setw()

stands for set width.
It is a manipulator in C++ used to set the width of output field.

Example:

```
#include <iostream>
#include <iomanip>
using namespace std;

int main () {
    cout << "Roll NO " << setw(10) <<
    "Name" << setw(15) << "Marks" << endl;
    cout << "1" << setw(13) << "Ali" << setw(13) << "Ali"
    << setw(12) << "81" << endl;
    cout << "2" << setw(13) << "Hori" << setw(12)
    << "92" << endl;
```

Roll No	Name	Marks
1	Ali	85
2	Haris	92
3	Sara	88

=== Code Execution Successful ===

endl;

It is a manipulator in C++ used to insert a new line in the output.

Example:

```
#include <iostream>
```

```
#include using namespace std;
```

```
int main () {
```

```
cout << "Haris Khan" << endl;
```

```
return 0;
```

```
}
```

setfill :

setfill is a manipulator in C++ used to fill the empty spaces that appear when you use setw().

setw() leaves blank spaces, but setfill allows you to fill those spaces with any character like ' , - , or #).

*****Haris Khan

-----Haris Khan

#####Haris Khan

=== Code Execution Successful ===

Haris khan

Welcome to C++ programming.

=== Code Execution Successful ===

Example setfill ();

```
#include <iostream>
#include <iomanip>
using namespace std;
```

```
int main ( );
```

```
cout << string name = "Haris Khan";
```

```
cout << setfill ('*') << setw(15) <<
name << endl;
```

```
cout << setfill ('-') << setw(15)
<< name << endl;
```

```
cout << setfill ('#') << setw(15)
<< name << endl;
```

```
return 0;
```

```
}
```


Set precision (n):

Set precision is a manipulator used to control how many digits are shown after the decimal point when printing floating-point numbers (like float or double).

Example:

```
#include <iostream>
#include <iomanip>
using namespace std;
```

```
int main () {
```

```
    cout << Name : Hon's Khan << endl;
    cout << "Original Percentage" << endl;
```

```
    cout << fixed << setprecision (2);
    cout <<
```

```
    return 0;
```

```
}
```


Student: Haris

Original Percentage: 86.7543

=== Code Execution Successful ===